

SIMULATION IDEAS

For each observation $i \in [n]$, denote by its tensor covariate $\mathcal{X}_i \in \mathbb{R}^{p_1 \times \dots \times p_D}$, one-way covariates $\mathbf{z}_i \in \mathbb{R}^{p_0}$, and two responses $y_{ij} \in \mathbb{R}$ where $j = 1, 2$ represents two eyes. Let $t \in [0, 1]$ be the observation time, $\mathcal{A}(t) = (\alpha_{j_1, \dots, j_d}(t)) \in \mathbb{R}^{p_1 \times \dots \times p_D}$ be the unknown tensor coefficient function, and $\boldsymbol{\beta} \in \mathbb{R}^{p_0}$ be the coefficients. The model is assumed to be

$$\begin{aligned} y_{ij} &= \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \\ &= \sum_{s=1}^S \left\langle \mathcal{X}_{ij}^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \end{aligned}$$

by a rank- R CP decomposition,

$$\begin{aligned} &\approx \sum_{s=1}^S \sum_{r=1}^R x_{ijrs}^* \left\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i) \right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \\ &= \langle \mathbf{X}_{ij}^*, \mathbf{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij}, \end{aligned}$$

where matrix $\mathbf{X}_{ij}^* := (x_{ijrs}^*) \in \mathbb{R}^{R \times S}$, and $\mathbf{A}(t) := (\alpha_{rs}(t)) \in \mathbb{R}^{R \times S}$ with $\alpha_{rs}(t) := \langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t) \rangle$. Within one patient, the eyes are correlated

$$\begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N} \left(\mathbf{0}, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right).$$

Remark. In the current Cases I–IV below, we use a simplified single-eye setting and do not consider within-subject correlation between the two eyes: no correlation in $(\epsilon_{i1}, \epsilon_{i2})^\top$. Hence, the DGP needs to be updated, and the method codes also needs to be updated accordingly.

Current Case I & II

The tensor size is set to be $D = 3$, $p_1 = p_2 = p_3 = 40$, and $S = 64$. We split the tensor by $S^{(1)} = S^{(2)} = S^{(3)} = 10$ and $p'_1 = p'_2 = p'_3 = 4$. Set the rank $R = 5$, then the whole-sample s -th sub-block tensor $\tilde{\mathcal{X}}^{(s)} \in \mathbb{R}^{n \times p'_1 \times p'_2 \times p'_3}$ can be generated by

$$\tilde{\mathcal{X}}^{(s)} = \sum_{r=1}^R \mathbf{x}_{rs}^* \circ \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$$

where $\mathbf{x}_{rs}^* \in \mathbb{R}^n$ is a Gaussian ensemble vector whose entries are iid from $\mathbf{N}(0, 1)$, and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient tensor function is set to be

$$\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$$

where $\alpha_{rs}(t)$ has 4 choices: $4\sqrt{\frac{rs}{RS}}(t-0.5)^2$, $\sqrt{\frac{rs}{RS}}t^{0.5}$, $1.75\sqrt{\frac{rs}{RS}}(\exp\{-(3t-1)^2\} + \exp\{-(4t-3)^2\} - 0.75)$, and $\sqrt{\frac{rs}{RS}}\sin(2\pi(t-0.5))$. The one-way vector $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is Gaussian ensemble matrix, and $\boldsymbol{\beta} = (3, 3)^\top$.

Remark. Case I is just the 2D version of Case II.

Current Case III & IV

For Case IV, the tensor size is set to be $D = 3$, $p_1 = p_2 = p_3 = 80$, and $S = 64$. We split the tensor by $S^{(1)} = S^{(2)} = S^{(3)} = 4$, so that $p'_1 = p'_2 = p'_3 = 20$. Set the rank $R = 20$, then the whole-sample s -th sub-block tensor $\tilde{\mathcal{X}}^{(s)} \in \mathbb{R}^{n \times p'_1 \times p'_2 \times p'_3}$ can be generated by $\tilde{\mathcal{X}}^{(s)} = \sum_{r=1}^R \mathbf{x}_{rs}^* \circ \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$, where $\mathbf{x}_{rs}^* \in \mathbb{R}^n$ is a Gaussian ensemble vector whose entries are iid from $N(0, 1)$, and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient tensor function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^R \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$, where

$$\alpha_{rs}(t) = \begin{cases} \sqrt{\frac{rs}{RS}} \sin(2\pi(t - 0.5)), & 1 \leq r \leq s \leq 10, \\ \sqrt{\frac{rs}{RS}}, & 1 \leq s < r \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

The one-way vector $\mathbf{Z} \in \mathbb{R}^{n \times p_0}$ is a Gaussian ensemble matrix, and $\boldsymbol{\beta} = (1, 1, 0, 0, 0)^\top$. The random error ϵ_i follows $N(0, 1)$. This case is designed for the high-dimensional setting, where $2RS + p_0 > n$, so the initial local linear kernel smoothing step is no longer feasible and the paper focuses on the penalized spline stage and the refined regression stage.

Remark. Case III is the 2D counterpart of Case IV, with spatial correlation across partitions being introduced. More specifically, for two partitions indexed by $s = (s^{(1)}, s^{(2)})$ and $s' = (s'^{(1)}, s'^{(2)})$, the corresponding latent scores are correlated according to an AR(1)-type structure with covariance

$$\rho^{|s^{(1)} - s'^{(1)}| + |s^{(2)} - s'^{(2)}|}.$$

Hence Case III is used to assess structure identification under spatially correlated tensor covariates, whereas Case IV is used to assess the same problem under a high-dimensional 3D setting.

Next Step 1: Two-Eye Update

This is the most urgent update. We can keep the signal structure in Cases I–IV unchanged, while extending the data-generating process to incorporate a two-eye structure: correlation in $(\epsilon_{i1}, \epsilon_{i2})^\top$. The corresponding estimation procedure also needs to be updated so that subject-level dependence between the two eyes is properly accounted for.

1 Other Idea: Generating Tensor Covariate

Currently, the tensor $\tilde{\mathcal{X}}$ is generated by tensor product of vector, like $\tilde{\mathcal{X}}^{(s)} = \sum_{r=1}^R \mathbf{x}_{rs}^* \circ \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$. To make the revised simulation more convincing, we can directly start generating covariate tensor $\mathcal{X}_i$ and the coefficient tensor $\mathcal{A}(t)$, so that the CP decomposition really has some information loss.

1.1 By Matrix Normal

In the 2D case, we may generate

$$\text{vec}(\mathcal{X}_i) \sim N(\mathbf{0}, \Sigma_2 \otimes \Sigma_1),$$

where $\mathcal{X}_i \in \mathbb{R}^{p_1 \times p_2}$ and each $\Sigma_d \in \mathbb{R}^{p_d \times p_d}$. This is the matrix-normal construction, and its higher-order extension leads to the tensor-normal model with separable covariance structure; see [1]. A simple choice is

$(\Sigma_d)_{uv} = \exp(-|u - v|/\ell_d)$, or a Matérn-type covariance if we want more flexible local regularity. In the 3D case, we may analogously generate

$$\text{vec}(\mathcal{X}_i) \sim N(\mathbf{0}, \Sigma_3 \otimes \Sigma_2 \otimes \Sigma_1),$$

for $\mathcal{X}_i \in \mathbb{R}^{p_1 \times p_2 \times p_3}$. This gives a directly generated image-like tensor with spatial correlation.

1.2 By Gaussian Process

We can also generally generate $\mathcal{X}_i$ as a spatial Gaussian random field or Gaussian process. In the 2D case, for pixel locations $s = (u, v)$ and $s' = (u', v')$, we may specify

$$\mathcal{X}_i(s) \sim \mathcal{GP}(0, C(s, s')),$$

where the covariance kernel can be chosen as

$$C(s, s') = \tau^2 \exp \left\{ -\frac{(u - u')^2}{\ell_1^2} - \frac{(v - v')^2}{\ell_2^2} \right\},$$

or more generally as a Matérn kernel. 3D case can mimic this idea. For a general review of Gaussian random field generation methods, see [2]. For an image-regression example where predictor images are simulated from Gaussian processes, see [3].

1.3 By Real Images Empirical Basis

Another possibility is to use real images to construct a more realistic simulation mechanism. Suppose we have preprocessed training images from the real-data study. Let $\bar{\mathcal{X}}$ be the sample mean image and let $\Psi_1, \dots, \Psi_{K_0}$ be the first several empirical principal component images extracted from the observed tensors. Then we may generate

$$\mathcal{X}_i = \bar{\mathcal{X}} + \sum_{k=1}^{K_0} \xi_{ik} \Psi_k + \mathcal{E}_i,$$

where the scores ξ_{ik} are sampled from a fitted distribution, for example $\xi_{ik} \sim N(0, \lambda_k)$ with λ_k equal to the estimated eigenvalue of the k -th principal component, and $\mathcal{E}_i$ is a residual spatial noise field. This empirical-PCA or eigenimage construction is close in spirit to simulation designs built from real neuroimaging scans; see [4].

2 Generating Tensor Coefficient Function

The coefficient tensor can be generated by $\mathcal{A}(t) = \sum_{k=1}^K g_k(t) \Phi_k$, where each Φ_k is a spatial template on the original image domain and each $g_k(t)$ controls how the strength of that region varies with t . For the 3D case, we may define three fixed spatial templates by

$$\begin{aligned} \Phi_1(u, v, w) &= \exp \left\{ -\frac{(u - d_{11})^2 + (v - d_{12})^2 + (w - d_{13})^2}{2h_1^2} \right\}, \\ \Phi_2(u, v, w) &= \exp \left\{ -\left(\frac{(u - d_{21})^2}{2a_1^2} + \frac{(v - d_{22})^2}{2a_2^2} + \frac{(w - d_{23})^2}{2a_3^2} \right) \right\}, \\ \Phi_3(u, v, w) &= -\exp \left\{ -\frac{(u - d_{31})^2 + (v - d_{32})^2 + (w - d_{33})^2}{2h_3^2} \right\}, \end{aligned}$$

where Φ_1 and Φ_3 are two localized spherical templates centered at (d_{11}, d_{12}, d_{13}) and (d_{31}, d_{32}, d_{33}) , and Φ_2 is an elongated ellipsoidal region centered at (d_{21}, d_{22}, d_{23}) . 2D case can be generated in the similar way.

But I am still considering how to add sparsity in this setting. Maybe we can still adopt the current $\mathcal{A}(t)$ generation method in Case IV.

References

- [1] Manceur, A. M. and Dutilleul, P. (2013). Maximum likelihood estimation for the tensor normal distribution: Algorithm, minimum sample size, and empirical bias and dispersion. *Journal of Computational and Applied Mathematics*, **239**, 37–49.
- [2] Liu, Y., Li, J., Sun, S., and Yu, B. (2019). Advances in Gaussian random field generation: a review. *Computational Geosciences*, **23**(5), 1011–1047.
- [3] Guo, C., Kang, J., and Johnson, T. D. (2022). A spatial Bayesian latent factor model for image-on-image regression. *Biometrics*, **78**(1), 72–84.
- [4] Goldsmith, J., Huang, L., and Crainiceanu, C. M. (2014). Smooth scalar-on-image regression via spatial Bayesian variable selection. *Journal of Computational and Graphical Statistics*, **23**(1), 46–64.